

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

A city employee will plant two types of bushes, azaleas and boxwoods, in a park. There will be no more than **164** total bushes planted, and the number of azaleas planted will be at most three times the number of boxwoods planted. Which of the following systems of inequalities best represents this situation, where *a* is the number of azaleas that will be planted, and *b* is the number of boxwoods that will be planted?

Answer

A. $a + b \geq 164$

$$3a \geq b$$

B. $a + b \geq 164$

$$a \leq 3b$$

C. $a + b \leq 164$

$$3a \geq b$$

D. $a + b \leq 164$

$$a \leq 3b$$

Correct Answer: D

Rationale

Choice D is correct. It’s given that a city employee will plant azaleas and boxwoods in a park, where *a* is the number of azaleas that will be planted and *b* is the number of boxwoods that will be planted. It’s also given that there will be no more than **164** total bushes planted, which can be represented by the inequality $a + b \leq 164$. It’s also given that the number of azaleas planted will be at most three times the number of boxwoods planted, which can be represented by the inequality $a \leq 3b$. Therefore, the system of inequalities containing $a + b \leq 164$ and $a \leq 3b$ best represents this situation.

Choice A is incorrect. The inequality $a + b \geq 164$ represents a situation where at least **164** total bushes will be planted, not that there will be no more than **164** total bushes planted. Also, the inequality $3a \geq b$ represents a situation where the number of boxwoods that will be planted is at most three times the number of azaleas that will be planted, not that the number of azaleas planted will be at most three times the number of boxwoods planted.

Choice B is incorrect. The inequality $a + b \geq 164$ represents a situation where at least **164** total bushes will be planted, not that there will be no more than **164** total bushes planted.

Choice C is incorrect. The inequality $3a \geq b$ represents a situation where the number of boxwoods that will be planted is at most three times the number of azaleas that will be planted, not that the number of azaleas planted will be at most three times the number of boxwoods planted.